Lecture - 13

① $u, v, w \in \mathbb{R}^7$
$\neq \{0\}$

They span a subspace, then what are the possible dimension

i) u, v, w are independent

$$\boxed{\dim(\text{span}(\underbrace{u, v, w}_{\text{basis}})) = 3}$$

ii) Two independent

$$\boxed{\dim = 2}$$

iii) Only one independent

$$\boxed{\dim = 1}$$

(2) i) 5×3 matrix U

rank = 3 in echelon form

$N(U) = \{0\}$ since no free vars

ii) 10×3 matrix $\begin{bmatrix} U \\ 2U \end{bmatrix}$

echelon form $\rightarrow$ $\begin{bmatrix} U \\ 0 \end{bmatrix}$

$\boxed{\text{rank} = 3}$

iii) $\begin{bmatrix} U & U \\ U & 0 \end{bmatrix}$

echelon form $\rightarrow$ $\begin{bmatrix} U & 0 \\ 0 & U \end{bmatrix}$

UI $\boxed{\text{rank} = 6}$

iv) $C = \begin{bmatrix} U & 0 \\ 0 & U \end{bmatrix}$

$\dim N(C^T) = ?$

$\dim N(C) = 0$ as no free var

$\dim (N(C^T)) = m - r = 10 - 6 = \underline{\underline{4}}$

③ $Ax = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}_{3\times 1}$

$$x = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

→ $A \rightarrow (3 \times n) \quad (n \times 1)$

$x \rightarrow (3 \times 1)$ from sol^{n}s

So, $\boxed{A \rightarrow 3 \times 3}$

2 free variables with spl sl's $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\boxed{\text{rank} = 1}$

$\boxed{\dim C(A^T) = 1}$

$\boxed{\dim N(A) = 2}$

Now, to construct matrix A

- $x_p = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$

$$A_{:1} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \qquad A_{:2} = \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}$$

$$A_{:3} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

→ $Ax = b$ is solvable when?

$b \rightarrow \lambda \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ b is in $C(A)$

④

i) A square matrix has null space of zero vector, then transpose has null space?

→ $A_{n \times n}$ $N(A) = \{0\}$

$\text{rank}(A) = n$

$N(A^T) = n - \text{rank}(A) = \boxed{0}$

$N(A^T) = \{0\}$

ii) 5×5 matrices →

Do invertible matrices form a subspace?

→ λA → invertible ✓

$\lambda_1 A + \lambda_2 B$ → not necessarily ⊗

Not a subspace.

iii) if $B^2 = 0$ then $B = 0$

$B \to$ square matrix

False , $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$B^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

iv) A system of n equations & n unknowns is solvable for every b if the columns are independent. True / False

$\boxed{n \times n}$ matrix $\boxed{\text{rank} = n}$

$C(A) = \mathbb{R}^n$

$\boxed{\text{Yes}}$ one solution

Basis $(C(A)) = I_n$

(5)

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

i) Basis for nullspace $N(B) \leq \mathbb{R}^4$

$$Bx = 0$$

$$\hookrightarrow \underbrace{E \begin{bmatrix} & \end{bmatrix}}_{I} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & \underbrace{0 \quad 0}_{\text{free vars}} \end{bmatrix} = 0$$

$$\dim(N(B)) = 2 \quad \underbrace{c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} + d \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}}_{\text{Basis}}$$

$N(CD) = N(D)$
if C is invertible!

ii) Solve

$$Bx = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{array}\right]$$

$$\rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$x_p \text{ [obvious one] } = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x = x_p + x_n = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} + d \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

(6)

i) If $n = m$, then the row space is equal to column space.

(Nopes) eg. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

The dimensions are same tho.

ii) The matrices A & $-A$ share the same 4 subspace

(Yes)

iii) If A & B have the same 4 subspaces then $A = \lambda B$.

Nope!! eg. Two square invertible matrices

then $\text{rank}(A) = \text{rank}(B)$

iv) If 2 rows of A are exchanged, which subspaces stay the same.

- Row space → same
- $N(A)$ → same

v) Why can't the vector $[1\ 2\ 3]$ can be a row and in the null space

$$\Rightarrow \quad v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Suppose $v \in N(A)$

$A \rightarrow n \times 3$

$$Av = 0$$

At least 1 for non

$$\begin{bmatrix} \vdots & \vdots & \vdots \\ | & | & | \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$2 \geq \text{rank}(A) \geq 0$$

$$A\, v = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$v \neq 0$ if $\underset{\text{is row}}{A_i} = v^T$

Then there would be one entry in $Av \rightarrow A_i v$ for which the value is $v \cdot v$ or $\|v\|^2$

$\|v\|^2 = 0$ only when all the components are zero which they are not so impossible.

Hence, the only overlap in row space & null space can be the zero vector!

$$\boxed{\underbrace{C(A^T) \cap N(A)}_{\text{"perpendicular spaces"}} = \{0\}}$$